S1.5 Ideal gases

The big picture

Guiding question(s)

How does the model of ideal gas behaviour help us to predict the behaviour of real gases?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Next time you are travelling in a car, think about how the behaviour of gases can affect your journey. From the air pressure in the tyres to the combustion of the hot fuel gases in the engines, gases play vital roles. You hopefully have not experienced another important function that relies upon understanding how gases behave – air bags. When a car is involved in a collision, sensors detect the change in forces and trigger the release of a gas into air bags that inflate in order to cushion and protect the passengers from hitting the structures of the car. The amazing thing is that all of this needs to occur between the start of the collision and the passengers being thrown forward in the car. Watch **Video 1** to learn how this is done and how it saves lives.

Video 1. The chemistry involved in air bags.

As shown in **Figure 1**, a crash sensor initiates a very fast chemical reaction that produces nitrogen gas. But what factors contribute to the filling of the airbag to the correct volume? This is all explained by the behaviour of gases which you will be exploring in this section.

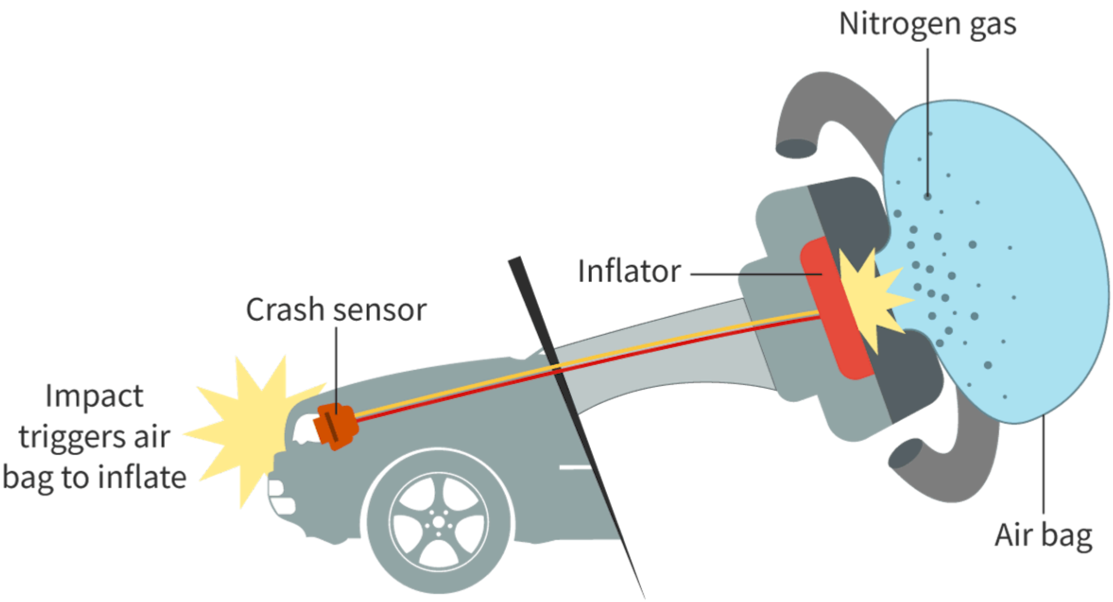

Figure 1. An air bag inflates when the crash sensor initiates the reaction that produces nitrogen to inflate the bag.

More information for figure 1

Concept

Structure is one the two main concepts in the DP chemistry course, with the other being reactivity. Structure in chemistry is concerned with the way in which the particles of a substance are arranged. In this section, you will learn the importance of how the structure of gases affect their physical properties.

Prior learning

Before you study this subtopic make sure that you understand the following:

- States of matter (see section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/))
- Mole conversions (see sections S1.4.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/relative-atomic-mass-and-relative-formula-mass-id-44261/) and S1.4.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-mass-m-id-44262/))
- Formula determination (see section S1.4.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/empirical-formula-and-molecular-formula-id-43572/))